

Bellman-Ford Algorithms

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1 Introduction

This document presents three versions of the Bellman-Ford algorithm:

- Using an edge list
- Using an adjacency matrix
- A custom matrix-based incremental update version

Algorithm 1 Bellman-Ford using Edge List

Require: Number of vertices V , edge list $edgeList$, source vertex $source$

Ensure: Shortest distances from $source$ or report negative cycle

```
1: Initialize array  $dist[0 \dots V-1] \leftarrow \infty$ 
2:  $dist[source] \leftarrow 0$ 
3: for  $i \leftarrow 1$  to  $V-1$  do
4:   for all edge  $(u, v, w)$  in  $edgeList$  do
5:     if  $dist[u] + w < dist[v]$  then
6:        $dist[v] \leftarrow dist[u] + w$ 
7:     end if
8:   end for
9: end for
10: for all edge  $(u, v, w)$  in  $edgeList$  do
11:   if  $dist[u] + w < dist[v]$  then
12:     return "Graph contains negative weight cycle"
13:   end if
14: end for
15: return  $dist$ 
```

Algorithm 2 Bellman-Ford using Adjacency Matrix

```
1: function BELLMANFORD-MATRIX( $V$ ,  $graph$ ,  $source$ )
2:    $dist \leftarrow$  array of size  $V$  initialized to  $\infty$ 
3:    $dist[source] \leftarrow 0$ 
4:   for  $i \leftarrow 1$  to  $V - 1$  do
5:     for  $u \leftarrow 0$  to  $V - 1$  do
6:       for  $v \leftarrow 0$  to  $V - 1$  do
7:         if  $graph[u][v] \neq \infty$  then
8:           if  $dist[u] + graph[u][v] < dist[v]$  then
9:              $dist[v] \leftarrow dist[u] + graph[u][v]$ 
10:          end if
11:        end if
12:      end for
13:    end for
14:  end for
15:  for  $u \leftarrow 0$  to  $V - 1$  do
16:    for  $v \leftarrow 0$  to  $V - 1$  do
17:      if  $graph[u][v] \neq \infty$  then
18:        if  $dist[u] + graph[u][v] < dist[v]$  then
19:          return "Graph contains negative weight cycle"
20:        end if
21:      end if
22:    end for
23:  end for
24:  return  $dist$ 
25: end function
```

Algorithm 3 Incremental Bellman-Ford Update using Adjacency Matrix Version 1

```
1: function BF( $G$ ,  $num\_vertices$ ,  $start$ ,  $Label$ ,  $Prev$ )
2:   if  $Label[start]$  is uninitialized then
3:     for  $i \leftarrow 0$  to  $num\_vertices - 1$  do
4:       if  $G[start][i] > 0$  then
5:          $Label[i] \leftarrow G[start][i]$ 
6:          $Prev[i] \leftarrow start$ 
7:       end if
8:        $Label[start] \leftarrow 0$ 
9:        $Prev[start] \leftarrow start$ 
10:    end for
11:    return
12:  end if
13:   $TempLabel \leftarrow$  copy of  $Label$ 
14:  for  $u \leftarrow 0$  to  $num\_vertices - 1$  do
15:    for  $v \leftarrow 0$  to  $num\_vertices - 1$  do
16:      if  $G[v][u] > 0$  and  $TempLabel[v]$  is initialized then
17:         $new\_distance \leftarrow TempLabel[v] + G[v][u]$ 
18:        if  $Label[u]$  is uninitialized or  $new\_distance < Label[u]$  then
19:           $Label[u] \leftarrow new\_distance$ 
20:           $Prev[u] \leftarrow v$ 
21:        end if
22:      end if
23:    end for
24:  end for
25: end function
```

Algorithm 4 Incremental Bellman-Ford Update using Adjacency Matrix Version 2

```
1: function BF( $G, n, start, value, prev$ )
2:   Initialize  $init \leftarrow \text{false}$ 
3:   Copy  $value$  to  $Label$ 
4:   for  $i \leftarrow 0$  to  $n - 1$  do
5:     if  $Label[i] = 0$  then
6:        $init \leftarrow \text{true}$ 
7:     end if
8:   end for
9:   if  $init = \text{false}$  then
10:     $Label[start] \leftarrow 0$ 
11:     $value[start] \leftarrow 0$ 
12:  end if
13:  for  $u \leftarrow 0$  to  $n - 1$  do
14:    if  $Label[u] \neq -1$  then
15:      for  $v \leftarrow 0$  to  $n - 1$  do
16:        if  $G[u][v] \neq 0$  then
17:          if  $Label[u] + G[u][v] < value[v]$  or  $value[v] = -1$  then
18:             $value[v] \leftarrow Label[u] + G[u][v]$ 
19:             $prev[v] \leftarrow u$ 
20:          end if
21:        end if
22:      end for
23:    end if
24:  end for
25: end function
```

Algorithm 5 Find Shortest Path using Bellman-Ford Algorithm

```
1: function BF_PATH( $G, n, start, end$ )
2:   Initialize  $Label[0 \dots n - 1] \leftarrow -1$ 
3:   Initialize  $Prev[0 \dots n - 1] \leftarrow -1$ 
4:   for  $i \leftarrow 1$  to  $n - 1$  do
5:     BF( $G, n, start, Label, Prev$ )
6:   end for
7:   Initialize empty list  $Path$ 
8:    $current \leftarrow end$ 
9:   while  $current \neq -1$  do
10:    Prepend  $current$  to  $Path$ 
11:     $current \leftarrow Prev[current]$ 
12:  end while
13:   Initialize  $result \leftarrow \text{empty string}$ 
14:   for  $i \leftarrow 0$  to  $|Path| - 1$  do
15:      $result \leftarrow result + \text{Char}(Path[i] + 'A')$ 
16:     if  $i < |Path| - 1$  then
17:        $result \leftarrow result + "$ "
18:     end if
19:   end for
20:   return  $result$ 
21: end function
```
